

Merge sort and insertion sort

Revise 360

Worksheet for the 360° experience: Merge and insertion works

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Merge and insertion works. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Trace merge sort splits and merges.
- Trace insertion sort.
- Compare the main sorting methods.

Starter: before you put the headset on

Merge sorted lists [2, 7] and [3, 5]. What is the sorted result?

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Key terminology

Merge

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Sorted prefix

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Divide the list

Turn right

Key facts in your own words:

Example to trace:

[7, 2, 6, 3]
[7, 2] [6, 3]
[7] [2] [6] [3]

Challenge: Why can splitting stop at one item?

Order these merge-sort stages.

Put these steps in order: Merge the pairs into the sorted list / Merge into sorted pairs / Split again into single items / Split the original list

Merge sort splits until...

2 Merge sorted groups

Turn right

Key facts in your own words:

Example to trace:

Left: 2, 7

Right: 3, 6

Take: 2, 3, 6

Append: 7

Result: 2, 3, 6, 7

Challenge: Which value is chosen after 2? Why?

Merge [1, 8] and [4, 6]. Result?

Which value is chosen first when merging [4,9] and [2,7]?

3 Read a merge segment

Turn around

Key facts in your own words:

Example to trace:

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if left[i] <= right[j]
    output left[i]
    i = i + 1
else: output right[j]
    j = j + 1
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Challenge: Why increment only the pointer used?

Left is exhausted but right has [9, 12]. What next?

After taking left[i], which pointer advances?

4 Insert into a prefix

Turn around

Key facts in your own words:

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Example to trace:

Start: 6, 2, 5, 1
Insert 2: 2, 6, 5, 1
Insert 5: 2, 5, 6, 1
Insert 1: 1, 2, 5, 6

Challenge: What is the sorted prefix after inserting 5?

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After inserting 3 into prefix [2, 7, 9], the prefix is...

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Insertion sort grows...

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5 Shift and insert

Turn left

Key facts in your own words:

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Example to trace:

Prefix: 2, 7, 9; key = 5
Shift 9 right
Shift 7 right
Place 5 after 2
Result: 2, 5, 7, 9

Challenge: Why save the key before shifting values?

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Insert key 4 into sorted prefix [1, 6, 8].

Put these steps in order: Place 4 after 1 / Shift 6 one place right / Shift 8 one place right / Save key 4

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When inserting 5 into [2,7,9], which values shift?

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6 Compare the methods

Turn left

Key facts in your own words:

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Example to trace:

Bubble: neighbour swaps

Insertion: prefix + key

Merge: split then merge

Typical merge uses extra space

Challenge: Which method can use few shifts on nearly sorted data?

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Match the method to its main approach.

Match each item to a description from the list below.

Split then merge sorted groups / Insert a key into a sorted prefix / Compare and swap neighbours

Item	Matching description
Bubble	
Insertion	
Merge	

Which method splits and merges groups?

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★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Insertion sort on [4, 1, 3, 2]: after inserting the first two keys, the list is...

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Which statement describes the usual school-level merge-sort implementation?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

How do you put a new playing card into a sorted hand?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Show the split and merge stages for [8, 3, 6, 1]. [4]

b) Show the list after each insertion for [5, 2, 4, 1]. [3]

c) Give one difference between merge and insertion sort. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Trace merge sort splits and merges.			
Trace insertion sort.			
Compare the main sorting methods.			